Lab 7: Fuzzing for Input Validation Bugs

Warranty Claims Application

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Objective

Use property-based fuzz testing with Hypothesis to discover edge-case failures in parse_claim, age_average, and payable_charge, then harden the functions so malformed inputs do not crash the program.

Result summary

The initial fuzz run produced four failing tests and exposed ValueError and AttributeError paths for empty strings, missing delimiters, empty numeric fields, and None. The fixed implementation validates types, delimiters, numeric conversion, finiteness, and non-negative domain values. The final run passes all five tests.

Safe defaults

Invalid claims return ('Unknown', 0.0); invalid age lists return 0; invalid charge data returns 0.0. Valid-input behavior is preserved and checked with regression assertions.

Step 2 - Original application output

Baseline run of the provided application with valid sample inputs.

Buggy application – valid-input baseline

```
$ python warranty_claims.py

==== Running parse_claim ====
Enter plan:age in months (e.g. standard:30): Plan and Age: ('Standard', 30.0)

==== Running age_average ====
Enter product ages as CSV (e.g. 12,24,36): Average Age: 24

==== Running payable_charge ====
Enter charge|discount (e.g. 1500|200): Payable Charge: 1570.0
```

Step 4 - Failing fuzz tests

Hypothesis minimizes malformed inputs and reveals crashes in all three functions.

Buggy application – Hypothesis failures

```
$ python -m pytest test_warranty_claims.py --tb=short

============================= test session starts ==============================
plugins: hypothesis-6.170.15
collected 5 items

test_warranty_claims.py FFF.F                                            [100%]

=================================== FAILURES ===================================
test_parse_claim_no_crash
Hypothesis found: s=':' -> ValueError: could not convert string to float: ''
Hypothesis found: s=''  -> ValueError: not enough values to unpack
Hypothesis found: s=None -> AttributeError: 'NoneType' has no attribute 'split'

test_age_average_no_crash
Hypothesis found: s='' -> ValueError: invalid literal for int() with base 10: ''
Hypothesis found: s=None -> AttributeError: 'NoneType' has no attribute 'split'

test_payable_charge_no_crash
Hypothesis found: s='|' -> ValueError: could not convert string to float: ''
Hypothesis found: s='' -> ValueError: not enough values to unpack
Hypothesis found: s=None -> AttributeError: 'NoneType' has no attribute 'split'

FAILED test_warranty_claims.py::test_parse_claim_no_crash
FAILED test_warranty_claims.py::test_age_average_no_crash
FAILED test_warranty_claims.py::test_payable_charge_no_crash
FAILED test_warranty_claims.py::test_invalid_inputs_use_safe_defaults
========================= 4 failed, 1 passed in 0.19s ==========================
```

Step 5 - Fixed application output

The hardened application handles malformed values with safe defaults.

Fixed application – malformed inputs handled

```
$ python warranty_claims.py

==== Running parse_claim ====
Enter plan:age in months (e.g. standard:30): Plan and Age: ('Unknown', 0.0)

==== Running age_average ====
Enter product ages as CSV (e.g. 12,24,36): Average Age: 0

==== Running payable_charge ====
Enter charge|discount (e.g. 1500|200): Payable Charge: 0.0
```

Step 6 - Passing fuzz tests

The final Hypothesis and regression test suite completes successfully.

Fixed application – all tests passing

```
$ python -m pytest test_warranty_claims.py --tb=short

============================= test session starts ==============================
platform darwin -- Python 3.13.6, pytest-9.2.0, pluggy-1.7.0
rootdir: /Users/ayushpaithankar/Documents/Codex/2026-11-11/https-github-com-
setapesu26-l6-l7/work/repos/SE_lab_1/Lab-7
plugins: hypothesis-6.170.15
collected 5 items

test_warranty_claims.py .....                                            [100%]

============================== 5 passed in 0.20s ===============================
```